

Ontology in Hybrid Intelligence: A Concise Literature Review

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ABSTRACT ORIGINAL

In the context of the constant evolution and proliferation of AI technology, hybrid intelligence is gaining popularity in reference to a balanced coexistence between human and artificial intelligence. The term has been extensively used over the past two decades to define models of intelligence involving more than one technology. This paper aims to provide (i) a concise and focused overview of the adoption of ontology in the broad context of hybrid intelligence regardless of its definition and (ii) a critical discussion on the possible role of ontology to reduce the gap between human and artificial intelligence within hybrid-intelligent systems, as well as (iii) the identification of possible future research directions in the field. Alongside the typical benefits provided by the effective use of ontologies at a conceptual level, the conducted analysis has highlighted a significant contribution of ontology to improving quality and accuracy, as well as a more specific role to enable extended interoperability, system engineering and explainable/transparent systems. Additionally, an application-oriented analysis has shown a significant role in present systems (70+% of cases) and, potentially, in future systems. However, despite the relatively consistent number of papers on the topic, a proper holistic discussion on the establishment of the next generation of hybrid-intelligent environments with a balanced co-existence of human and artificial intelligence is fundamentally missed in the literature. Last but not the least, there is currently a relatively low explicit focus on automatic reasoning and inference in hybrid-intelligent systems. © 2024 by the author.